

# Jaypee Institute of Information Technology, Noida



## Data Science & Data Analytics PBL REPORT

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***“Data Driven Analysis and Predictive  
Modelling of VLSI Design Metrics”***

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## **Abstract**

With the increasing complexity of Very Large Scale Integration (VLSI) systems, analyzing design metrics has become essential for improving chip performance, power efficiency, and manufacturability. This project presents a data-driven analysis and predictive modelling approach for evaluating VLSI design metrics obtained from the OpenLane RTL-to-GDSII flow using the Sky130 Process Design Kit (PDK).

The project focuses on the analysis of an 8-bit Arithmetic Logic Unit (ALU) design. Design metrics such as cell count, routing utilization, wire length, timing parameters, power consumption, and physical layout statistics were extracted from OpenLane-generated reports and converted into structured datasets for analysis.

Exploratory Data Analysis (EDA) techniques were applied to visualize and interpret the behavior of physical, routing, timing, and power-related parameters. Various plots including power breakdown, routing layer utilization, timing analysis, and physical cell distribution were generated to understand the characteristics of the design.

In addition to analytics, predictive modelling using Linear Regression was demonstrated to estimate power consumption and layout area based on cell count. Since the project contained only a single real design sample, a simulated dataset based on realistic VLSI scaling trends was generated to demonstrate the machine learning methodology.

The analysis showed that switching power dominated total power consumption, routing primarily utilized lower metal layers, and the design achieved zero DRC, LVS, and timing violations. The predictive models showed strong linear trends for power estimation while also highlighting the limitations of simplified models in capturing real-world physical design overhead.

The project successfully demonstrates the integration of VLSI design and data analytics concepts using Python-based data science tools.

# 1. Introduction

## 1.1 Background

Modern semiconductor systems contain millions to billions of transistors integrated on a single chip. As VLSI designs become more complex, analyzing design metrics such as area, timing, power consumption, routing congestion, and manufacturability becomes increasingly important.

Traditional VLSI workflows generate a large amount of design data during synthesis, floorplanning, placement, routing, and signoff stages. These reports contain valuable information that can be analyzed using data analytics techniques to better understand design behavior and improve optimization.

With the rise of data science and machine learning, data-driven approaches are increasingly being explored in Electronic Design Automation (EDA) for design prediction, optimization, and trend analysis.

## 1.2 Objective of the Project

The main objectives of this project are:

- To extract and analyze VLSI design metrics from OpenLane reports.
- To transform raw design metrics into structured datasets suitable for analytics.
- To perform Exploratory Data Analysis (EDA) on physical, timing, routing, and power metrics.
- To visualize important design characteristics using Python libraries.
- To demonstrate predictive modelling using machine learning techniques.
- To study the relationship between cell count, power consumption, and layout area.

## 1.3 Problem Statement

VLSI design tools generate complex design reports that are difficult to interpret manually. There is a need for a structured and data-driven approach to analyze these metrics and derive meaningful insights regarding design quality, power efficiency, timing performance, and physical implementation.

This project addresses this problem by combining VLSI design analysis with data analytics and predictive modelling techniques.

## 2. Tools and Technologies Used

| Tool/Technology | Purpose                          |
|-----------------|----------------------------------|
| OpenLane        | RTL-to-GDSII VLSI design flow    |
| Sky130 PDK      | Open-source process design kit   |
| Python          | Data analysis and visualization  |
| Pandas          | Data handling and transformation |
| NumPy           | Numerical operations             |
| Matplotlib      | Data visualization               |
| Seaborn         | Statistical visualization        |
| Scikit-learn    | Machine learning models          |
| Google Colab    | Development environment          |

## **3. Methodology**

### **3.1 Design Flow**

The project workflow consists of the following stages:

1. RTL Design of 8-bit ALU
2. OpenLane RTL-to-GDSII Flow Execution
3. Extraction of Design Metrics
4. Dataset Creation and Cleaning
5. Exploratory Data Analysis (EDA)
6. Predictive Modelling using Linear Regression
7. Interpretation of Results

### **3.2 Dataset Collection**

The design metrics were collected from the OpenLane-generated metrics.csv file. The dataset contained information related to:

- Physical design metrics
- Routing statistics
- Timing reports
- Power consumption
- Cell distribution
- Signoff verification

The original dataset was available in wide-column format.

### **3.3 Data Transformation**

The raw metrics were transformed into a structured long-format dataset using Python and Pandas. Important metrics such as:

- Cell count
- Wire length
- Number of vias
- Timing slack
- Power components
- Area utilization

were extracted and categorized into:

- Physical Metrics
- Routing Metrics
- Timing Metrics
- Power Metrics

This transformation improved readability and simplified data visualization.

### **3.4 Exploratory Data Analysis (EDA)**

EDA was performed to understand the behavior and relationships among different design parameters.

The following visualizations were generated:

- Overall metrics overview
- Power breakdown analysis
- Standard cell distribution
- Routing layer utilization
- Physical cell breakdown
- Timing analysis summary
- Design signoff scorecard

### **3.5 Predictive Modelling**

Machine learning techniques were demonstrated using Linear Regression.

Since the project contained only one real design sample, simulated datasets based on realistic VLSI scaling trends were generated to demonstrate predictive modelling.

Two models were implemented:

1. Cell Count vs Power Consumption
2. Cell Count vs Core Area

These models helped illustrate how data analytics can be integrated into VLSI design analysis.

## 4. Exploratory Data Analysis

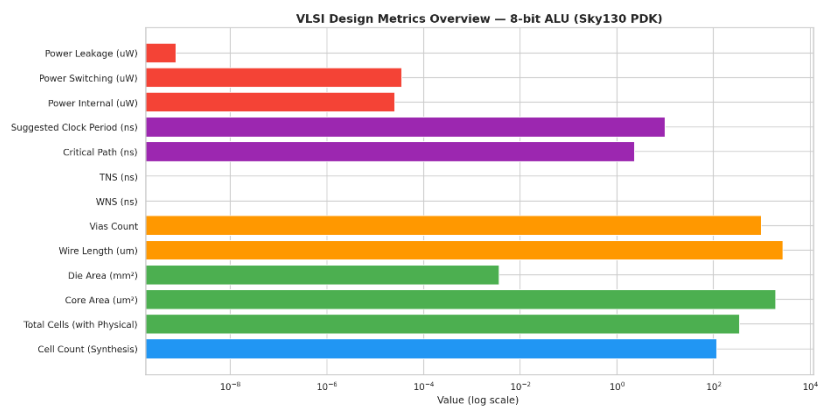
### 4.1 Metrics Overview

A log-scaled metrics overview plot was generated to compare parameters with significantly different magnitudes.

The analysis showed that:

- Wire length and physical area were among the largest metrics.
- Leakage power was extremely small compared to switching and internal power.
- Timing metrics remained within acceptable operating limits.

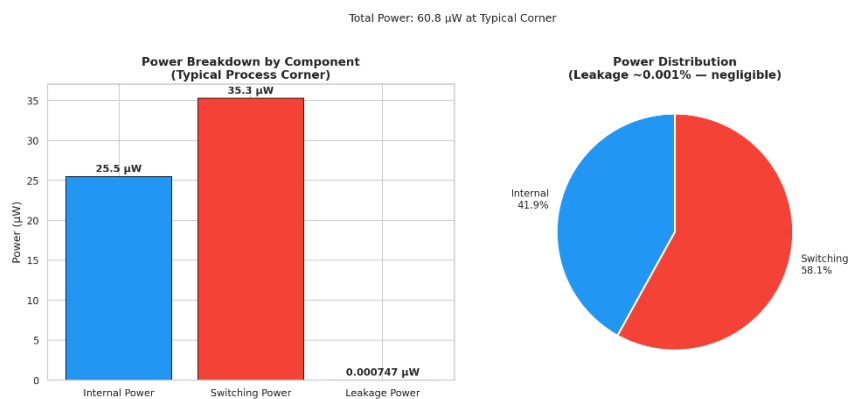
The use of logarithmic scaling improved the visibility of smaller metrics.



### 4.2 Power Breakdown Analysis

Power consumption was divided into:

- Internal Power
- Switching Power
- Leakage Power



## Observations

- Switching power contributed approximately 58.1% of total power.
- Internal power contributed approximately 41.9%.
- Leakage power was negligible.

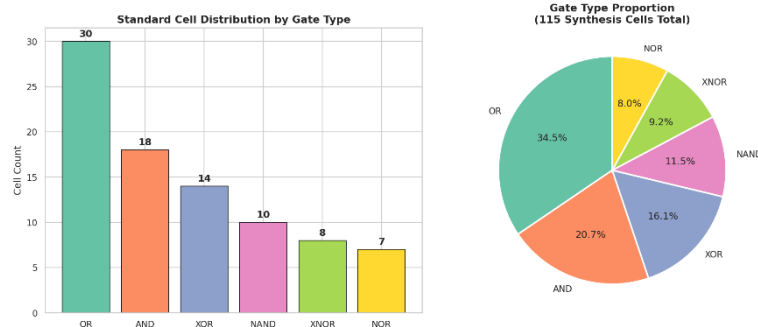
This behavior is expected in combinational CMOS logic where dynamic switching dominates power consumption.

### 4.3 Standard Cell Distribution

The distribution of logic gates used in the ALU was analyzed.

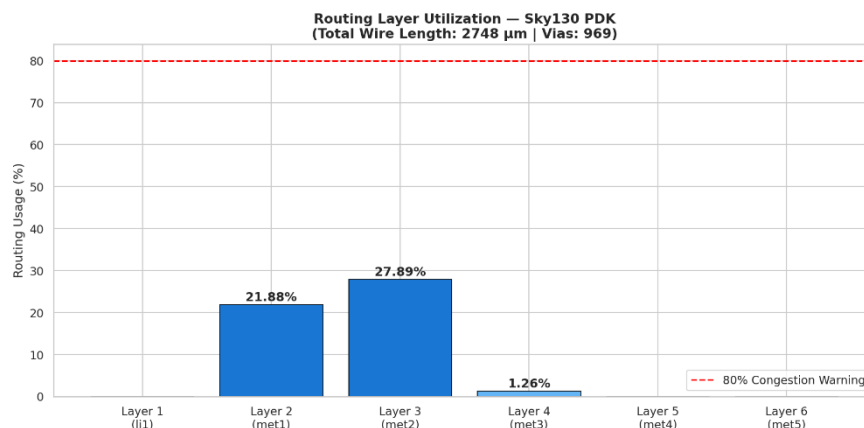
#### Observations

- OR gates formed the largest group.
- AND, XOR, and NAND gates were also significantly utilized.
- The distribution matched the arithmetic and logical operations implemented in the ALU.



### 4.4 Routing Layer Utilization

Routing layer usage across different metal layers was visualized.



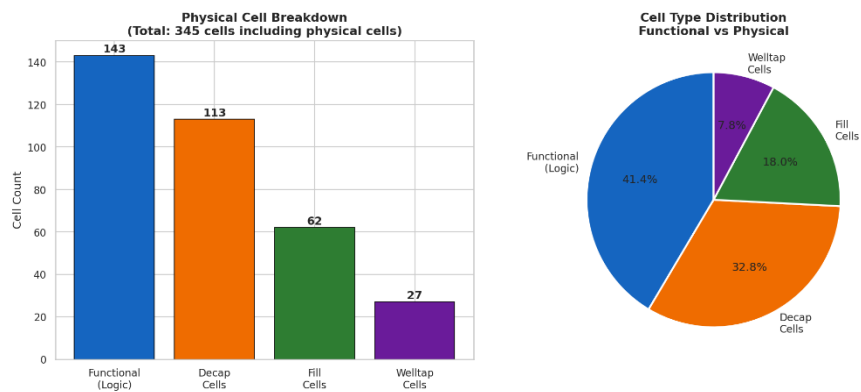
## Observations

- Most routing occurred on met1 and met2 layers.
- No routing congestion was observed.
- All routing layers remained well below the congestion threshold.

## 4.5 Physical Cell Breakdown

The design included:

- Functional logic cells
- Decap cells
- Fill cells
- Welltap cells



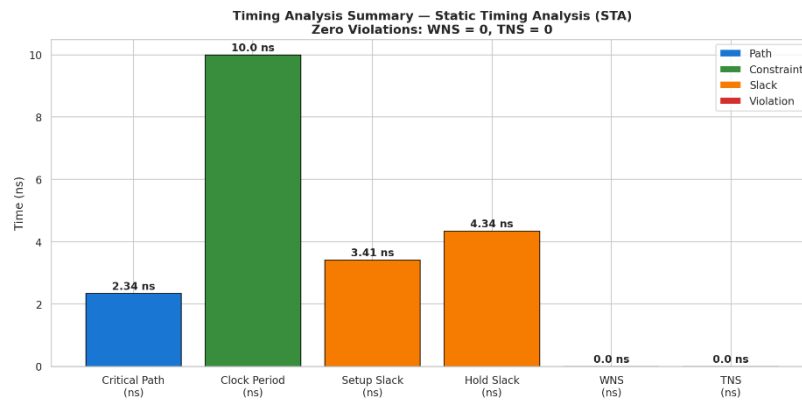
## Observations

- A large number of physical support cells were present.
- Decap and fill cells contributed significantly to overall area.
- Physical overhead played an important role in final layout area.

## 4.6 Timing Analysis

Timing analysis focused on:

- Critical path delay
- Clock period
- Setup slack
- WNS and TNS



## Observations

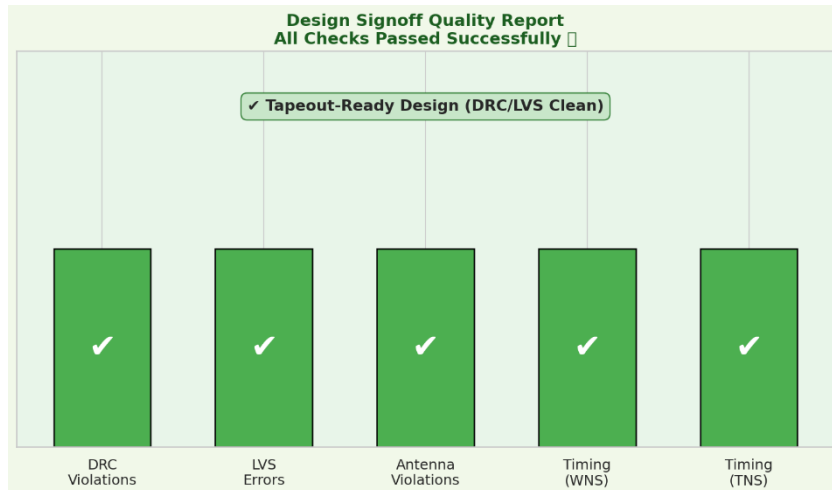
- Critical path delay was 2.34 ns.
- Clock period was 10 ns.
- Positive setup slack indicated successful timing closure.
- WNS and TNS were zero, indicating no timing violations.

## 4.7 Design Quality Analysis

The design passed all signoff checks:

- DRC Violations = 0
- LVS Errors = 0
- Antenna Violations = 0
- Timing Violations = 0

This confirms that the generated GDSII layout is manufacturable and tapeout-ready.



## 5. Machine Learning Analysis

### 5.1 Power Prediction Model

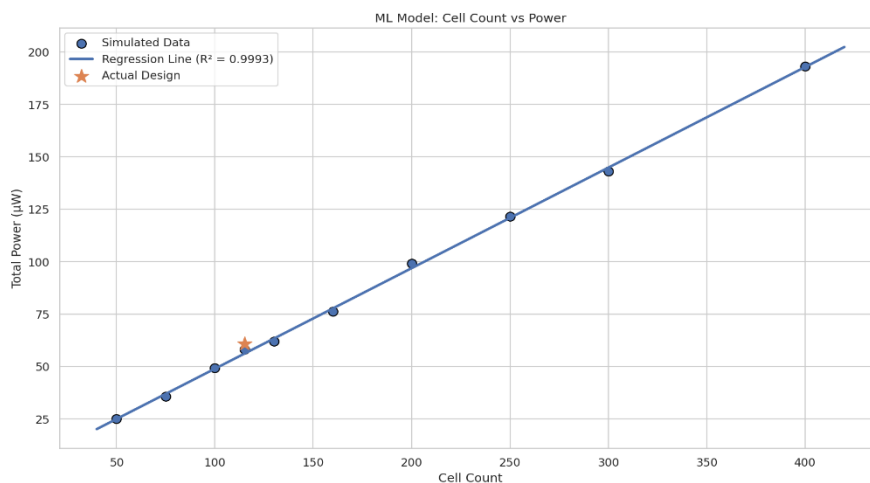
A Linear Regression model was trained using simulated VLSI scaling data.

#### Model Objective

Predict total power consumption based on cell count.

#### Results

- $R^2$  Score  $\approx 0.999$
- Predicted power for 115 cells  $\approx 56.13 \mu\text{W}$
- Actual power  $\approx 60.8 \mu\text{W}$



## Interpretation

The model demonstrated strong linear scaling between cell count and power consumption.

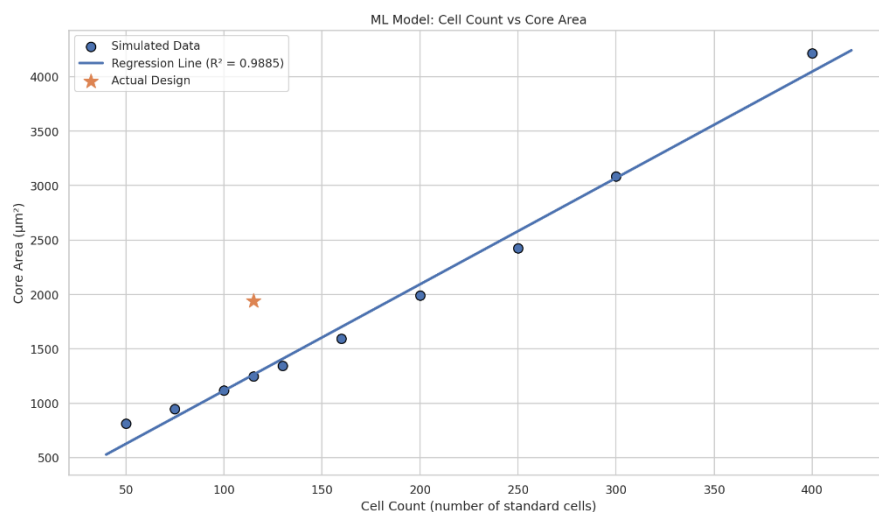
The prediction error was relatively small, indicating that power approximately scales with the number of active cells.

## 5.2 Area Prediction Model

A second Linear Regression model was implemented to predict layout area.

### Results

- $R^2$  Score  $\approx 0.988$
- Predicted area  $\approx 1262 \mu\text{m}^2$
- Actual area  $\approx 1941.86 \mu\text{m}^2$



## Interpretation

Although the model captured the general scaling trend, it significantly underestimated the actual area.

This occurred because:

- Physical routing overhead
- Decap cells
- Fill cells
- Welltap structures
- Spacing rules

are not captured in a simple cell-count-based model.

This highlights an important real-world limitation of simplified machine learning models in VLSI physical design.

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## 6. Results and Discussion

The project successfully demonstrated the use of data analytics techniques for analyzing VLSI design metrics.

### Key Findings

- The OpenLane flow completed successfully.
- The design achieved zero DRC, LVS, and timing violations.
- Switching power dominated total power consumption.
- Routing was concentrated mainly on lower metal layers.
- The ML-based power prediction model showed strong linear behavior.
- The area prediction model highlighted the impact of physical design overhead.

The project also demonstrated how data science tools can be integrated into EDA workflows for improved analysis and insight generation.

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## 7. Future Scope

Future improvements may include:

- Running multiple OpenLane design configurations.
  - Creating larger real-world datasets.
  - Applying advanced ML algorithms.
  - Predicting congestion and timing violations.
  - Automating report generation.
  - Integrating AI-driven design optimization.
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## **8. Conclusion**

This project demonstrated a successful integration of VLSI design analysis and data analytics techniques using Python-based tools.

Metrics obtained from the OpenLane RTL-to-GDSII flow were transformed into structured datasets and analyzed using exploratory data analysis methods. Multiple visualizations provided insights into routing behavior, power distribution, timing performance, and physical design overhead.

Predictive modelling using Linear Regression illustrated how machine learning can be applied to estimate VLSI design characteristics. While the power prediction model showed strong accuracy, the area prediction model highlighted the importance of physical design complexity beyond simple logical abstraction.

Overall, the project successfully showcased how data-driven methodologies can improve understanding and interpretation of VLSI design metrics.

## References

1. OpenLane Documentation
2. SkyWater SKY130 PDK Documentation
3. Scikit-learn Documentation
4. Pandas Documentation
5. NumPy Documentation
6. Matplotlib Documentation
7. S.-M. Kang and Y. Leblebici, CMOS Digital Integrated Circuits: Analysis and Design